University of Illinois
ECE 310 **Profs. Do, Shomorony, Snyder**

Midterm Exam 1

7:00-9:00pm, Wednesday, October 2, 2024

Name: _____

Section: 9:00 AM 12:00 PM 3:00 PM

NetID: _____

Score: _____

Problem	Pts.	Score
1	12	
2	12	
3	8	
4	10	
5	15	
6	15	
7	18	
8	10	
Total	100	

Instructions

- You may not use any books, calculators, or notes other than one handwritten two-sided sheet of 8.5" x 11" paper.
- Show all your work to receive full credit for your answers.
- When you are asked to "calculate", "determine", or "find", this means providing closed-form expressions (i.e., without summation or integration signs).
- Neatness counts. If we are unable to read your work, we cannot grade it.
- Turn in your entire booklet once you are finished. No extra booklet or papers will be considered.

Netid: _____

(12 Pts.)

1. Select **True** or **False** to each of the following statements:

   (a) For a discrete-time system with impulse response $h[n]$, the system output $y[n]$ to any input signal $x[n]$ is always determined using $h[n]$. **True/False**

   (b) The impulse response $h[n]$ of an unstable LTI system will always be an unbounded sequence. **True/False**

   (c) Two BIBO stable LTI systems connected in parallel will always form a BIBO stable LTI system. **True/False**

   (d) An LCCDE system can only have a finite number of poles. **True/False**

   (e) An LTI system given by $y[n] = y[n-3] + x[n]$ will have three distinct poles in its transfer function. **True/False**

   (f) An LTI system with transfer function $H(z) = \frac{1}{1-0.5z^{-1}} + \frac{1}{1-2z^{-1}}$ must **not** be BIBO stable. **True/False**

Netid: _____

(12 Pts.)

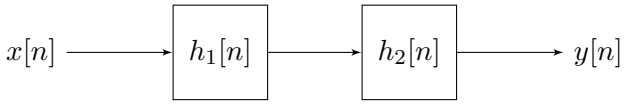
2. For each of the systems with input $x[n]$ and output $y[n]$ shown in the table, indicate by "**yes**" or "**no**" whether the properties indicated apply to the system.

	Linear	Shift-Invariant	Causal	Stable
$y[n] = x[n]\ x[n+1]$				
$y[n] = \frac{1}{\lvert n\rvert+1} x[n]$				
$y[n] = \sin(x[n]) + x[0]$				

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(8 Pts.)

3. Consider a system composed of two LTI systems connected in series.



The input $x[n] = \{\underset{\uparrow}{1}, -1\}$ produces output $y[n] = \{4, \underset{\uparrow}{-2}, -2\}$. We also know that $h_2[n] = \{\underset{\uparrow}{2}, 1\}$.

(a) Determine $h_1[n]$.

(b) Determine the overall system's impulse response $h[n]$ such that $y[n] = x[n] * h[n]$. Is this overall system causal? Justify your reasoning.

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(10 Pts.)

4. For each of the following parts, compute the convolution $x[n] * h[n]$ for the given sequences.

   (a) $x[n] = \{1, -3, \underset{\uparrow}{2}, -1, 4\},\ h[n] = \{2, \underset{\uparrow}{0}, -1\}$

   (b) $x[n] = e^{-n}u[n],\ h[n] = (n+1)(u[n] - u[n-3])$

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(15 Pts.)

5. For each of the following discrete-time signals, find its z-transform and the corresponding region of convergence.

   (a) $x[n] = (n+1)u[n-1]$.

   (b) $x[n] = u[n-1]u[3-n]$.

   (c) $x[n] = 3^{-n}u[n] + 3^{n}u[-n]$.

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(15 Pts.)

6. A causal LTI system produces the output $y[n] = \{1, 1/2, 0, 0, 0, 0, \ldots\}_{n=0}^{\infty}$ when excited by the input signal $x[n] = \{1, 1/3, 0, 0, 0, 0, \ldots\}_{n=0}^{\infty}$.

   (a) Determine the transfer function of the system.

   (b) Determine the impulse response of the system.

   (c) Express the system in the form of an LCCDE.

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(18 Pts.)

7. A causal LTI system has transfer function given by

$$H(z) = \frac{1}{(1 - \frac{1}{2}z^{-1})(1 - 2z^{-1})}.$$

(a) Sketch the ROC of $H(z)$ and clearly indicate the location of the poles.

(b) Find the impulse response $h[n]$.

(c) Suppose the input to this system is right-sided and has z-transform

$$X(z) = \frac{z^{-1} - 2z^{-2}}{1 - \frac{1}{2}z^{-1}}.$$

Find the z-transform $Y(z)$ of the output of the system and its ROC. Check for any pole-zero cancellations that may occur.

(d) Determine the the output signal $y[n]$ corresponding to $Y(z)$ in part (c). Is $y[n]$ bounded?

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(10 Pts.)

8. (a) Suppose we have a **non-causal** LTI system described by:

$$y[n] + \alpha y[n-1] = x[n] + 3x[n-1],$$

where α is a finite, real-valued constant. For what value(s) of α is this system BIBO stable?

(b) Suppose we have a **causal** LTI system described by:

$$y[n] - \frac{5}{2}y[n-1] + y[n-2] = x[n] + \beta x[n-1],$$

where β is a finite, real-valued constant. For what value(s) of β is this system BIBO stable?